Retail Promotion Effectiveness

Campaign, coupon, category, and household-level performance analysis

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Technical workflow: Python, EDA, ETL/ELT, PostgreSQL, SQL, and Power BI

This document evaluates how campaigns, promotions, and coupons relate to sales performance, purchase frequency, household response, and promotional efficiency. The analysis integrates retail transactions, promotional exposure, redemptions, product attributes, and household data into a single analytical framework designed for decision-ready reporting.

Executive summary

This project evaluates promotional effectiveness by integrating transactions, campaigns, coupons, coupon redemptions, products, promotions, and household attributes into a unified analytical layer. The objective is to distinguish scalable performance signals from low-support or inconsistent results and translate them into decision-ready recommendations.

- Campaigns delivered the strongest commercial scale in the current output. Campaign 18 led aggregate net sales and household reach, while Campaign 27 showed the strongest directional incremental lift among campaigns with a valid pre-period baseline.
- Category response was concentrated in a small set of categories. SOFT DRINKS combined the most relevant balance of scale and positive response, while FLUID MILK PRODUCTS and BEEF showed persistent underperformance.
- Promotional response varied across household groups. Age 55–64, 1 Adult No Kids households, and selected income bands produced the most favorable directional signals in the segment-level analysis.
- Exposed households materially outperformed non-exposed households in average sales and trip frequency, although that comparison should be treated as descriptive rather than causal.
- Coupon redemptions were concentrated in a few campaigns and were largely one-time behaviors, suggesting that coupons acted more as activation mechanisms than as repeat-purchase drivers during the observed windows.
- The workflow combines Python-based data cleaning and EDA, ETL/ELT loading into PostgreSQL, SQL KPI modeling, data quality validation, and Power BI reporting.

Business context

In retail environments, promotions, coupons, and campaign activations are recurring commercial levers used to drive sell-through, increase basket value, and influence customer behavior. Their impact, however, is rarely uniform. The same promotional action can perform very differently depending on category dynamics, product mix, timing, store conditions, or the characteristics of the households being reached. In that setting, a reliable analytical foundation is essential to separate initiatives that generate incremental value from those that merely shift demand or dilute margin without producing sustainable business impact.

This project builds an analytical solution to evaluate promotion and campaign effectiveness using transactional data, promotional exposure records, and coupon redemption behavior. The work combines data modeling, EDA, quality validation, ETL/ELT loading into PostgreSQL, SQL analysis, and dashboard design to understand household response, identify the categories most responsive to promotional activity, and uncover opportunities to improve commercial strategy. Rather than simply reporting historical sales, the goal is to turn fragmented data into a decision-ready framework that supports promotional investment decisions, demand planning, and commercial prioritization.

Objectives

- Measure the impact of promotions, campaigns, and coupons on purchasing behavior across sales, frequency, basket value, and household response.
- Identify which categories, products, or customer groups are the most promotion-sensitive and generate stronger commercial outcomes.

- Distinguish between initiatives that create genuine performance uplift and those with limited, inconsistent, or low-value results.
- Build a reliable analytical foundation to monitor outcomes, compare initiatives, and support commercial decisions with clear and traceable metrics.

Analytical questions

1. Which promotions, campaigns, or coupons are associated with stronger outcomes in sales, purchase frequency, and basket value?
2. Which categories and products appear to be the most responsive to promotional activity?
3. How does promotional response vary across different households or customer segments?
4. What differences can be observed between promotion-exposed households and non-exposed households?
5. Which initiatives appear to drive meaningful incremental performance, and which ones show limited or inconsistent results?
6. Where are the clearest opportunities to improve promotional investment across categories, products, or customer groups?
7. What coupon redemption patterns help explain stronger or weaker commercial response?
8. How can a reliable analytical foundation be built to continuously monitor promotional effectiveness and support commercial decision-making?

Data sources and model

The analytical foundation of this project is built on The Complete Journey, a household-level retail dataset that captures one year of grocery purchasing activity for 2,469 households from a supermarket chain. In addition to purchase history, the source includes information on campaigns, coupons, coupon redemptions, promotions, products, and demographic attributes for a subset of households, making it suitable for analyzing the relationship between commercial exposure and purchasing behavior from a relational and business-oriented perspective. According to the official package documentation, the source contains eight integrated datasets. Two of them are distributed as samples within the package (transactions_sample and promotions_sample), while the full versions of transactions and promotions are available from the source repository.

From a data model perspective, the source combines reference tables and event-level tables. products, campaign_descriptions, and demographics provide descriptive context; campaigns, coupons, and coupon_redemptions support campaign and redemption logic; and transactions together with promotions form the transactional core used to measure sales, promotional exposure, and household response. The project uses the full tables when scale and completeness materially affect interpretation and relies on sampled exposure data only where the package structure or analytical support requires it. Whenever sampled promotional exposure drives a result, that limitation is stated explicitly in the corresponding section.

Dataset summary

Dataset	Dataset Description	Main Columns
campaigns	Links households to the campaigns in which they participated.	campaign_id, household_id
campaign_descriptions	Contains campaign metadata, including campaign type and active date range.	campaign_id, campaign_type, start_date, end_date

Dataset	Dataset Description	Main Columns
coupons	Lists coupons associated with campaigns and the products for which they are valid.	coupon_upc, product_id, campaign_id
coupon_redemptions	Records coupon redemptions by household, campaign, and date.	household_id, coupon_upc, campaign_id, redemption_date
demographics	Includes demographic information for a subset of households.	household_id, age, income, home_ownership, marital_status, household_size, household_comp, kid_count
products	Provides product- and manufacturer-level descriptive attributes.	product_id, manufacturer_id, department, brand, product_category, product_type, package_size
promotions / promotions_sample	Describes product-level promotional presence in stores and mailers by week.	product_id, store_id, display_location, mailer_location, week
transactions / transactions_sample	Records household purchases at the transaction and product level.	household_id, store_id, basket_id, product_id, quantity, sales_value, retail_disc, coupon_disc, coupon_match_disc, week, transaction_timestamp

Note. The table above provides a functional summary of the source. The official documentation contains more detailed field-level definitions, especially for discounts, prices, promotional placement, and campaign logic.

Methodological approach

The workflow follows an Ask-Prepare-Process-Analyze-Share-Act structure adapted to a retail promotional-effectiveness use case. The approach combines business understanding, Python-based preparation and EDA, ETL/ELT loading into PostgreSQL, SQL analysis, data quality control, visualization, and executive communication. The aim is to move from raw relational inputs to a consistent analytical foundation and then to decision-ready interpretation.

The methodology is organized into eight connected stages, matching the execution flow of the project. It begins with business question definition and source review, continues through data modeling, cleaning, standardization, exploratory analysis, and quality validation, and then moves into analytical table creation, KPI development, and reporting. This structure keeps the project coherent across business framing, engineering discipline, and analytical interpretation.

Project workflow summary

1. Define business questions and the KPI set required to evaluate promotional effectiveness.
2. Review source tables, granularity, keys, and dataset relationships.

3. Design the relational model connecting campaigns, promotions, products, coupons, redemptions, households, and transactions.
4. Clean, standardize, and validate the source data in Python; convert data types; and perform EDA across numeric, categorical, and temporal variables.
5. Load the processed tables into PostgreSQL through an ETL/ELT workflow and create indexes to support scalable analytical querying.
6. Build business-facing analytical tables for sales, exposure, redemption, response, and category or campaign performance.
7. Develop SQL queries and complementary Python analysis to measure KPIs, compare groups, and test directional performance patterns.
8. Translate the output into an executive document and a Power BI dashboard oriented to decision-making.

Data engineering and quality

Data preparation and quality assurance

This stage focused on loading the source tables into Python and performing an initial structural review of records, data types, and column completeness. The inspection confirmed the presence of the core entities in the analytical model — campaigns, products, coupons, redemptions, promotions, households, and transactions — and identified the main adjustments required before analysis, primarily related to variable typing and temporal formatting.

Selected columns originally loaded as float were validated as identifier fields with no actual decimal values. Based on that assessment, product_id in products and coupons, as well as basket_id in transactions_sample, were converted to integer format to ensure consistency across joins and downstream transformations. Likewise, start_date, end_date, redemption_date, and transaction_timestamp were converted to appropriate date or datetime formats to support temporal consistency across tables and analyses.

The quality review did not reveal material structural issues in the categorical fields. No duplicated categories were found as a result of inconsistent formatting, capitalization, or residual spacing. Several variables were intentionally preserved as text because of their analytical meaning, including age, income, and kid_count in demographics, as well as package_size in products, since they represent ranges or measurement-based labels rather than continuous numeric values.

An EDA and anomaly-screening step was also performed across numeric, temporal, and categorical variables to identify unexpected ranges, outliers, and potential inconsistencies. Overall, no issues were found that required corrective action at this stage. The main point of attention was quantity in transactions_sample, which showed a strong concentration at low values and a long upper tail with extreme observations. Although clearly non-normal, the pattern was considered plausible in a transactional retail context and did not justify immediate intervention without stronger evidence of data error.

Area	Action taken	Outcome
Initial loading	Reviewed table structure, record counts, and base data types in Python.	Confirmed structural consistency across the core entities.
ID standardization	Converted product_id and basket_id to integer where appropriate.	Consistent typing for joins and downstream analysis.

Area	Action taken	Outcome
Date handling	Converted campaign, redemption, and transaction fields to date or datetime.	Improved temporal consistency across tables and analyses.
Categorical review	Compared raw and normalized labels to detect formatting duplicates.	No clear category duplication caused by formatting issues.
Missing values	Reviewed nulls in descriptive dimensions and flagged them for selective handling.	Preserved analytical context without forcing unnecessary imputation.
EDA / anomaly screening	Reviewed distributions, ranges, and sparse patterns across key variables.	No critical corrections required before modeling.

PostgreSQL data loading

After the initial preparation and validation steps, the processed tables were loaded into PostgreSQL to support relational querying, entity-level joins, and downstream analysis directly in SQL. This was done by connecting Python to PostgreSQL through SQLAlchemy and psycopg and then exporting the cleaned pandas DataFrames into database tables while preserving the analytical structure defined during the preparation stage.

The loading process included campaign, campaign description, product, coupon, coupon redemption, promotion, demographic, and transaction tables. Existing table versions were replaced when necessary to maintain a consistent analytical database. Once the tables were loaded, indexes were created on the main analytical keys — including campaign, product, household, store, week, and basket identifiers — to improve the performance of joins, filters, and KPI aggregation across the model.

In practical terms, this step marked the transition from a Python-based preparation environment to a relational analytical layer in PostgreSQL, leaving the data ready to answer the project's business questions through SQL and to support a more scalable reporting workflow.

Analytical framework

This analysis is structured on a consolidated SQL data model designed to measure promotional effectiveness from both a commercial and a customer-response perspective. By integrating transactions, promotions, campaigns, coupons, coupon redemptions, products, and household attributes, the framework enables a consistent evaluation of how promotional activity translates into measurable business outcomes across products, categories, and customer segments.

The framework uses comparative KPI analysis rather than a single evaluation lens. Some sections compare promoted versus non-promoted performance at the category level, others evaluate households relative to their own pre-campaign baseline, and others assess descriptive differences between exposed and non-exposed groups. Because each analytical question requires a different valid population, household counts are not expected to be identical across all tables. Campaign participant counts, baseline-qualified households, coupon-eligible households, and exposed-household cohorts are related but not interchangeable denominators.

To reduce ambiguity in interpretation, the core KPIs used throughout the document are defined below.

Metric	Definition used in this document
Net Sales	Sum of sales_value at the grain required by the analysis.
Avg Basket Value	Net sales divided by trips within the relevant cohort or initiative.
Purchase Frequency	Trips per household in aggregate views or weekly trips per household in incremental analyses.
Sales Lift %	Relative change in sales between comparison groups or time windows.
Trip Lift %	Relative change in trips between comparison groups or time windows.
Unit Lift %	Relative change in units sold between comparison groups or time windows.
Responder Rate	Share of households showing a positive response under the defined comparison logic.
Exposure Rate	Share of observed category activity associated with promotional exposure in the sampled category-opportunity analysis.
Redemption Rate	Redeemed households divided by eligible households for the coupon-campaign combination.

Results and business recommendations

Answers to the questions

Campaigns show the strongest commercial performance

The strongest commercial results in the first comparison were concentrated at the campaign level rather than in isolated promotion placements or redeemed coupons. Campaign 18 (Type A) delivered the highest net sales at 27,181.76, supported by 1,034 participating households, 5,229 trips, and a purchase frequency of 5.06. Campaigns 13 and 8 also ranked strongly, showing that the most consistent performance was associated with initiatives that combined broader household reach with repeated purchasing activity rather than isolated promotional events.

From an analytical perspective, this suggests that campaigns were the most scalable mechanism within the current sample, while promotion placements and redeemed coupons generated materially smaller volumes and were supported by thinner observation bases. SQL was used to integrate transactions with campaign, promotion, and coupon-redemption records and to calculate core KPIs such as net sales, trips, households reached, average basket value, and purchase frequency.

Initiative	Households	Trips	Net Sales	Avg Basket Value	Purchase Frequency
Campaign 18 | Type A	1,034	5,229	27,181.76	5.20	5.06
Campaign 13 | Type A	956	4,261	21,097.33	4.95	4.46
Campaign 8 | Type A	967	4,302	20,780.20	4.83	4.45

Initiative	Households	Trips	Net Sales	Avg Basket Value	Purchase Frequency
Campaign 14 | Type C	221	1,588	8,805.95	5.55	7.19

Note. Promotion-placement and redeemed-coupon results were retained for exploratory comparison, but their lower sample size and less stable KPI behavior make campaign-level findings more reliable for decision-making. Household counts in this table reflect participating households within the aggregate campaign view and should not be read as directly equivalent to the baseline-qualified populations used in later sections.

Promotional response by category

Promotional responsiveness was concentrated in a small number of categories, with only six meeting the minimum support threshold after the analysis was adjusted to account for sparse promotional exposure in the sample. Under a promoted-versus-base comparison, CRACKERS/MISC BKD FD and SOFT DRINKS showed the strongest positive sales response. CRACKERS/MISC BKD FD ranked first with a sales_lift_pct of 23.9% and a modest positive unit_lift_pct of 3.2%, while SOFT DRINKS delivered the most commercially relevant combination of scale and performance, with 10 promotional trips, 42.19 in promotional sales, and a sales_lift_pct of 15.1%.

The remaining categories showed weaker or negative total category response. YOGURT, BEEF, FLUID MILK PRODUCTS, and COLD CEREAL posted negative category-level sales lift under this specific comparison logic. In SQL, this question was addressed by comparing promotional versus non-promotional performance at the category level and calculating KPIs such as promotional sales, base sales, promotional sales per trip, base sales per trip, sales lift, and unit lift. Because promotional exposure was limited in the sampled data, the findings should be interpreted as directional rather than definitive.

Category	Promo Trips	Promo HHs	Promo Sales	Promo Sales/Trip	Base Sales/Trip	Sales Lift %	Unit Lift %	Interpretation
CRACKERS/MISC BKD FD	5	3	12.01	2.40	2.24	23.9%	3.2%	Positive but low-support signal
SOFT DRINKS	10	5	42.19	4.22	2.93	15.1%	-16.5%	Strongest scalable positive signal
YOGURT	5	4	8.79	1.76	1.41	-20.5%	-63.8%	Weak response under total-lift view
BEEF	7	6	38.13	5.45	6.44	-41.2%	-27.6%	Negative promotional response
FLUID MILK PRODUCTS	15	12	26.23	1.75	2.45	-56.6%	-53.5%	Clear underperformance
COLD CEREAL	5	5	9.16	1.83	2.97	-60.9%	-43.4%	Clear underperformance

Note. Category responsiveness in this section is based on total category lift under a promoted-versus-base comparison. It should therefore be interpreted separately from the later opportunity matrix, which uses sales per trip lift and exposure rate rather than total lift alone.

Promotional response by household segment

The segment-level analysis shows that promotional response varied across household groups, although the strength of that response was not uniform. Among age groups, households aged 55–64 delivered the clearest positive signal, with a sales_lift_pct of 5.1%, a trip_lift_pct of 8.5%, and a unit_lift_pct of 31.8%, alongside the highest responder rate among age segments at 55.3%. In household composition, 1 Adult No Kids showed the most favorable overall pattern, with modest positive sales lift and stronger unit growth. By income, the most positive sales response was observed in 100–124K households, while Under 15K also showed a meaningful increase in both sales and unit volume.

From a technical standpoint, the SQL logic was designed to produce a more meaningful segmentation view than a simple promoted-versus-non-promoted row comparison. The query first measured household-level performance during campaign windows relative to each household's own pre-campaign baseline and then aggregated the results by demographic segment. Null values were excluded only within the relevant dimension being analyzed, which improved segment clarity without unnecessarily removing households from other valid segment views.

Segment Dimension	Segment	Households	Sales Lift %	Trip Lift %	Unit Lift %	Responder Rate	Interpretation
Age	55–64	59	5.1%	8.5%	31.8%	55.3%	Strongest age-based response
Household composition	1 Adult No Kids	248	2.5%	-1.3%	14.6%	50.5%	Most favorable composition segment
Income	100–124K	32	14.2%	-0.1%	20.7%	55.4%	Strongest sales-based income segment
Income	Under 15K	59	7.9%	-2.9%	29.4%	54.1%	Positive directional signal

Note. Null demographic values were excluded only within the segment dimension being analyzed, preserving as much valid household information as possible across the broader segmentation view. Results from very small segments should be interpreted cautiously.

Exposed vs. non-exposed households

A clear performance gap can be observed between exposed and non-exposed households. Exposed households generated substantially higher average sales per household (132.37 vs. 32.98) and median sales per household (101.36 vs. 20.99), while also showing a much higher purchase frequency, with 26.58 average trips per household compared with 6.95 for non-exposed households. Average basket value was only moderately higher for exposed households (4.88 vs. 4.64), suggesting that the main difference between the two groups was driven more by shopping frequency than by basket size.

In analytical terms, SQL was used to classify households into exposed and non-exposed cohorts based on campaign participation, promotional exposure, or coupon redemption and then compare their household-level KPIs. This produced a descriptive cohort view across sales, trips, median household sales, basket value, and unit volume. The result indicates that exposed households were materially more engaged and commercially valuable, although the comparison should not be interpreted as purely causal promotional impact.

Cohort	Households	Avg Sales/HH	Median Sales/HH	Avg Trips/HH	Avg Basket Value	Interpretation
Exposed	1,565	132.37	101.36	26.58	4.88	Higher-value, higher-frequency households
Non-exposed	812	32.98	20.99	6.95	4.64	Lower-value, less active households

Note. This cohort comparison is descriptive, not causal. More active households may also be more likely to receive or respond to promotional activity.

Incremental performance by campaign

The incremental analysis showed that most campaigns delivered either limited or mixed results, with only a small number showing clearly positive directional signals. Among campaigns with a valid pre-campaign baseline, Campaign 27 (Type A) stood out as the strongest positive candidate, generating a sales_lift_pct of 32.9%, a trip_lift_pct of 27.0%, and a responder_rate of 54.3% across 361 assigned households. Campaign 18 (Type A) showed a more moderate lift pattern but the broadest scale, with 1,133 households and a 7.3% sales lift, making it the strongest scalable signal rather than the strongest intensity signal.

Most other campaigns produced weak gains, flat performance, or negative lift relative to their own pre-campaign baseline. SQL was used to compare household-level performance during each campaign window against an equally sized pre-campaign period, normalizing results to weekly sales and weekly trips per household. Because this section requires a valid pre-period baseline, its household counts are not directly comparable to the aggregate campaign counts shown earlier.

Campaign	Type	Households	During Sales/HH/Week	Pre Sales/HH/Week	Sales Lift %	Trip Lift %	Responder Rate	Interpretation
27	Type A	361	2.88	2.17	32.9%	27.0%	54.3%	Strongest positive directional signal
18	Type A	1,133	3.00	2.79	7.3%	4.5%	50.0%	Strongest scalable positive signal
4	Type B	81	3.47	3.17	9.6%	-3.4%	53.1%	Moderate / mixed
6	Type C	65	5.33	4.92	8.4%	-4.0%	52.3%	Moderate / mixed
20	Type C	244	2.28	4.08	-44.1%	-47.4%	23.4%	Clear underperformance
23	Type B	183	0.64	5.02	-87.2%	-88.4%	2.7%	Severe underperformance

Note. Campaigns with zero pre-campaign sales may appear in the broader output but should not drive business interpretation because incremental lift cannot be reliably computed when the baseline is zero.

Promotional investment opportunities by category

The clearest category-level opportunities for promotional investment were concentrated in SOFT DRINKS and YOGURT, both of which combined low promotional exposure with positive sales efficiency lift. SOFT DRINKS showed the strongest improvement, with a sales_per_trip_lift_pct of 44.21% at a very low exposure rate, while YOGURT also delivered a positive lift of 24.52% under similarly limited exposure. Under this lens, both categories appear to offer upside for selective scaling because current support remains low while observed return per trip is directionally favorable.

This section should be interpreted differently from the earlier category-response table. The earlier view measures total category lift under a promoted-versus-base comparison, whereas the opportunity matrix combines exposure rate with sales per trip lift. A category can therefore appear weak in total lift yet still show positive efficiency under limited exposure. YOGURT fits that pattern: it was weak in aggregate category lift but positive in per-trip efficiency, which suggests a selective test-and-validate opportunity rather than an unconditional scale recommendation.

Department	Category	Exposure Rate	Sales per Trip Lift %	Recommended Action	Interpretation
GROCERY	SOFT DRINKS	0.39%	44.21%	Scale investment	Strongest upside opportunity
GROCERY	YOGURT	0.66%	24.52%	Selective scale / validate	Positive efficiency under low exposure
GROCERY	CRACKERS/MISC BKD FD	0.50%	7.07%	Selective test / monitor	Positive but modest signal
MEAT	BEEF	0.74%	-15.38%	Reduce or redesign	Underperforming category
GROCERY	FLUID MILK PRODUCTS	1.01%	-28.66%	Reduce or redesign	Clear negative return
GROCERY	COLD CEREAL	0.72%	-38.39%	Selective test / monitor	Weak performance, limited support

Note. Promotional exposure in the sampled data was limited across many categories, which reduced analytical support for broader category comparison. These results should therefore be treated as directional prioritization signals rather than final investment decisions.

Coupon redemption patterns

Coupon redemption activity was concentrated in a small number of campaigns, with Campaign 18 and Campaign 13 standing out as the largest drivers of coupon engagement across multiple product categories. The highest observed redemption rates were generally within the 8% to 13% range, indicating that coupon take-up was present but still limited relative to the full eligible household base. Repeat redemption remained near zero in most campaign-category combinations, suggesting that coupon response was typically one-time rather than sustained.

From a technical perspective, SQL was used to identify eligible households, match redemptions to campaign windows, and calculate coupon KPIs such as redemption rate, repeat redemption rate, redemption lag, sales per redeemer, and trips per redeemer. Post-redemption metrics were intentionally measured on same-product

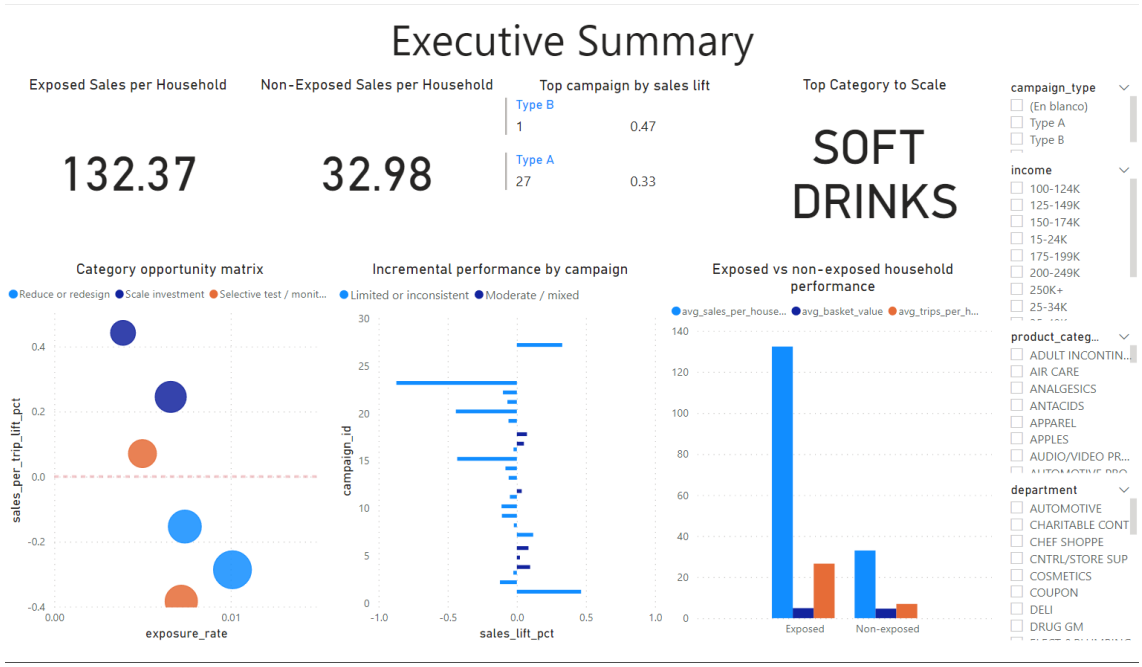
purchases after the first redemption date, which makes the analysis conservative and more likely to produce low post-redemption values in sparse transaction windows. The clearest conclusion is that coupons supported initial engagement more than repeat purchasing during the observed campaign periods.

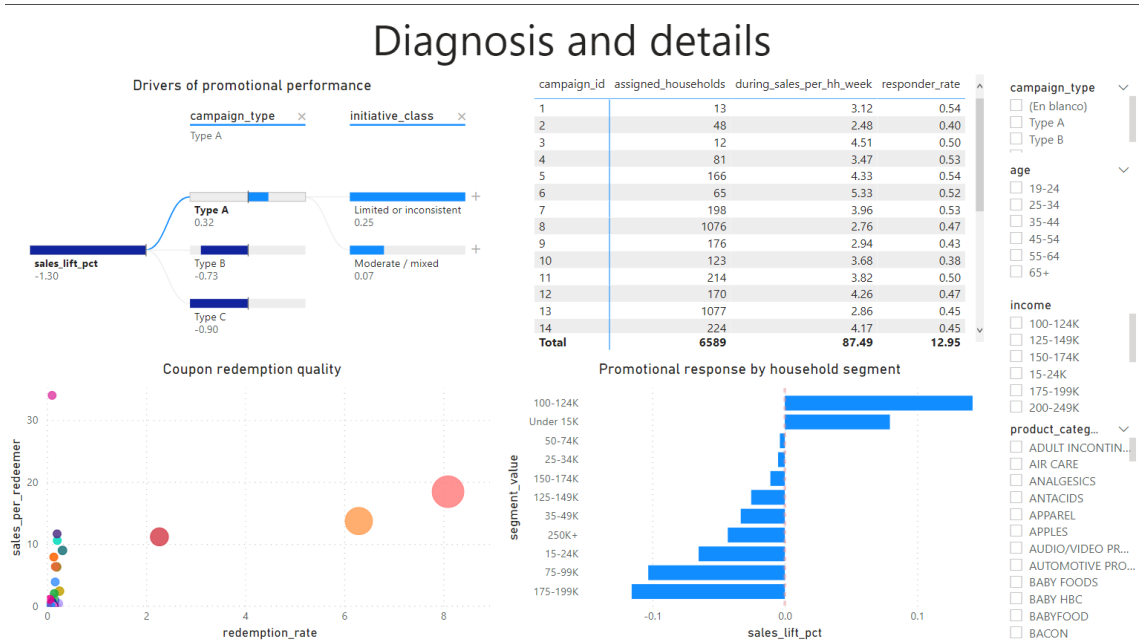
Campaign	Category	Eligible HHs	Redeemed HHs	Redemption Rate	Repeat Redemption Rate	Avg Redemption Lag (Days)	Interpretation
18	SEAFOOD - MISC	1,133	121	10.68%	0.00%	18.25	Broad reach, one-time redemption pattern
18	TURKEY	1,133	113	9.97%	0.00%	18.10	Strong reach, limited repeat behavior
13	SEAFOOD - MISC	1,077	89	8.26%	1.12%	16.99	Positive engagement, low repetition
19	DIAPERS & DISPOSABLES	130	10	7.69%	10.00%	16.15	Small base, relatively stronger repeat signal
15	CAT FOOD	17	2	11.76%	0.00%	11.64	High rate but very small base

Note. Repeat redemption remained limited in most campaign-category combinations. Coupon-eligible household counts in this table are specific to coupon logic and should not be compared directly with campaign totals or exposed-household cohorts from other sections.

Dashboard

A two-page Power BI dashboard was developed and connected to the PostgreSQL analytical database built during the ETL and SQL analysis stages. This ensured that the reporting layer remained directly aligned with the structured analytical outputs used throughout the case study. The first page, Executive Summary, was designed to highlight the most relevant business findings through a compact set of KPI cards and high-level visuals focused on campaign performance, category opportunities, and the gap between exposed and non-exposed households. The second page, Diagnostic and Detail, was built to support deeper exploration of campaign, coupon, and household-segment results through more analytical visuals and interactive filtering. Overall, the dashboard was structured to complement the case study by making the main findings faster to interpret, easier to compare, and more actionable from a business perspective





Limitations and next steps

- Promotional exposure in the sampled data was relatively sparse, so some promoted-versus-base comparisons should be interpreted with caution when support levels were low.
- Several findings are better understood as directional rather than definitive, particularly at the category and coupon level where low exposure introduced more noise and less stable lift estimates.
- The exposed-versus-non-exposed household comparison is descriptive, not causal, and some campaign results cannot be interpreted reliably when the pre-campaign baseline is zero or too thin.
- Extending the analysis to the full transactions and promotions tables would improve coverage and strengthen category- and product-level evaluation.
- Future work should incorporate more rigorous incremental approaches, such as matched controls, refined pre/post validation, and stronger retention measurement.
- The current PostgreSQL and Power BI solution provides a solid monitoring foundation, but it should continue to evolve toward more continuous performance tracking and more robust promotional decision support.